

MAHJONG: An Augmented Reality Rehabilitation Serious Game for Motor-Cognitive Dual-Task Training in Older Adults

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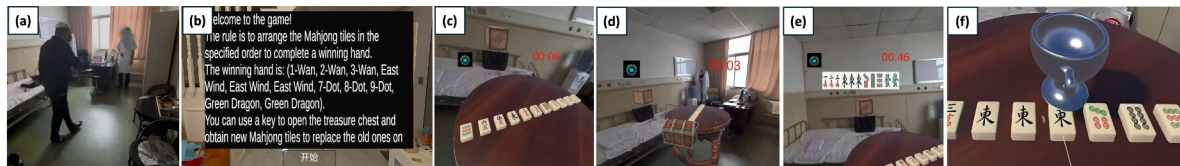


Figure 1: Game design overview: (a) environment mapping, (b) instructional prompt, (c) initial 11-tile Mahjong hand on the virtual table, (d) virtual chest revealing a new tile, (e) target-hand hint display, and (f) trophy-based reward mechanism reflecting hint usage.

ABSTRACT

Designing engaging rehabilitation training for older adults with augmented reality (AR) remains challenging because of age-related physical, cognitive, and technological barriers. This study presents an AR serious game for motor–cognitive dual-task training, informed by Self-Determination Theory and flow theory, and based on the cultural familiarity of Mahjong to reduce learning barriers. A feasibility study with 30 hospitalized older adults with mild motor and/or cognitive impairments showed high perceived usability, strong intrinsic motivation, reduced anxiety, and improved acceptance. These results suggest that culturally contextualized AR game design can effectively increase engagement in rehabilitation for older adults.

Index Terms: Augmented reality, rehabilitation, gamification, motor-cognitive training

1 INTRODUCTION

Recent technological advances are shaping rehabilitation training, and augmented and virtual reality (AR/VR) is attracting growing interest due to their immersive characteristics and the potential to improve motivation [3]. However, despite the widespread use of immersion and gamification in AR-based rehabilitation, the diversity of physical and cognitive abilities between target populations remains unaddressed, potentially limiting the generalizability of existing findings to older clinical settings [4]. Older adults often face physical constraints, slower cognitive processing, and limited familiarity with immersive technologies, which can reduce engagement and training effectiveness.

Although gamification has shown promise to improve motivation among older adults, the role of cultural factors in the design of rehabilitation games has received comparatively little attention. Existing work indicates that integrating familiar activities and culturally meaningful practices can increase engagement and acceptance of interactive rehabilitation technologies [5]. In addition, traditional games such as Mahjong have been shown to support cognitive training in aging populations, including attention, memory,

and executive function [7]. Motivated by these observations, this study introduces an AR-based Mahjong rehabilitation game within a motor–cognitive dual-task framework, utilizing interactive AR mechanics and culturally grounded serious game design to promote engagement and sustained participation.

2 MATERIALS AND METHODS

The AR game was developed using Unity¹ and deployed on a Meta Quest 3 head-mounted display². The game follows a structured interaction sequence. At launch, a rules interface displays the target winning hand (e.g., two triplets and one pair), allowing participants unfamiliar with Mahjong to interpret the objective as a pattern-matching task. Upon selecting Start, the instructions disappear and 11 Mahjong tiles are randomly dealt onto a virtual table. During gameplay, players interact with virtual “mystery chests” that are randomly distributed within the physical environment. For each revealed tile, players decide whether to retain or discard it in order to assemble the target hand.

The game was designed to adhere to a motor–cognitive dual-task training paradigm, which requires the concurrent execution of two different tasks (Fig. 1). The motor task involves physical spatial exploration: participants navigate their real-world environment (a hospital room) to approach and open virtual mystery chests that are randomly positioned in space. Opening a chest reveals a Mahjong tile. The cognitive task is twofold. First, players must memorize the target winning hand that defines the gameplay objective. Second, they perform a simultaneous continuous backward-counting task (e.g. subtracting by one from a given number) to impose sustained demands on working memory and attentional resources.

The game mechanics were informed by Self-Determination Theory (SDT) [2] and flow theory. The Relatedness is promoted through the Mahjong theme, which uses cultural familiarity to establish an immediate emotional connection and reduce the perceived distance from the training task. The Autonomy is supported by flexible pace, allowing participants to progress at their own speed, while an optional timer introduces time pressure to balance challenge and control, a key condition for flow. The Competence is reinforced through an on-demand hint function: participants may press a “Hint” button at any time to re-display the target hand for

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¹Unity Editor, version 2022.3.62f1 (Unity Technologies, San Francisco, CA, USA)

²Meta Quest 3 head-mounted display (Meta Platforms, Inc., Menlo Park, CA, USA)

10 seconds. To encourage mastery while accommodating individual impairments, the reward system (trophies) scales rewards and penalties based on hint usage, promoting a sense of achievement without discouraging necessary assistance.

This study recruited 30 inpatients from Cangzhou People's Hospital (17 men, 13 women; age $\mu = 61.6$, $\sigma = 6.1$). Inclusion criteria were a diagnosis of mild cognitive and/or motor impairment and the ability to walk independently. All participants were technology novices with no prior exposure to AR or VR.

The experimental procedure consisted of three phases. First, participants completed a 15-minute training session that included safety instructions and hands-on practice with the headset and controller. Second, they engaged in a 20-minute AR gameplay session while simultaneously performing the continuous backward-counting task. Finally, participants completed standardized questionnaires, including the System Usability Scale (SUS; 5-point Likert scale) [1] and the Intrinsic Motivation Inventory (IMI; 7-point Likert scale) [6]. The IMI comprised six subscales: interest/enjoyment, perceived competence, effort/importance, pressure/tension, perceived choice, and value/utility. A semi-structured interview followed to collect qualitative feedback on the Mahjong theme and operational challenges.

3 RESULTS AND DISCUSSION

The participants rated the game highly on the SUS (Fig. 2a), with a mean score of 79.0 ($\sigma = 9.25$), exceeding the standard benchmark of 68 and indicating excellent usability. The highest item-level scores reflected the willingness of the participants to use the game frequently ($\mu = 4.53$) and their perception that the system's functions were well integrated ($\mu = 4.83$).

Regarding intrinsic motivation, the overall mean score in the IMI (Fig. 2b) was relatively high ($\mu = 5.21$), suggesting a generally positive motivation response. In particular, the subscales of interest/enjoyment ($\mu = 6.01$) and value/usefulness ($\mu = 6.10$) received high scores, indicating that the game was engaging and perceived as a meaningful training activity. The pressure/tension subscale showed the lowest mean score ($\mu = 4.14$), suggesting that the experience did not induce excessive anxiety.

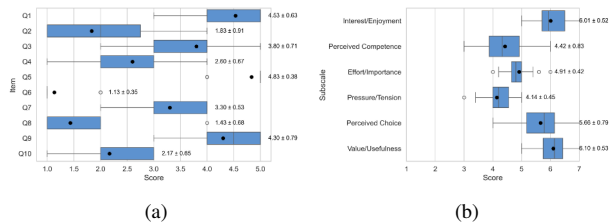


Figure 2: (a) Overall SUS results for ten items and (b) overall IMI results for six subscales.

Analysis of in-game performance revealed that the think-aloud backward-counting task imposed substantial cognitive demands. Most of the participants ($n = 22$) required five or more reminders or corrections from the researcher to complete the task successfully, indicating the challenges in maintaining concurrent cognitive processing. This finding highlights the high cognitive load associated with the dual-task design for older participants. However, qualitative feedback indicated that 21 participants perceived the AR game as more immersive and exploratory than conventional rehabilitation exercises, which contributed to increased interest and engagement.

Meanwhile, the results highlight several limitations. The frequent need for researcher intervention during the backward-counting task suggests that the imposed cognitive load may be too

demanding for this population, potentially exceeding their capacity for sustained attention and concurrent task management. In addition, repeated comments regarding headset weight and controller complexity point to the need for more inclusive hardware solutions or alternative interaction modalities, such as gesture-based input.

Although Mahjong-based design provides an effective cultural foundation, the observed cognitive and usability challenges indicate that the current implementation requires further refinement to support sustained use and therapeutic compliance. Moreover, since Mahjong is culturally familiar primarily in regions such as China and Southeast Asia, its limited recognition elsewhere may restrict immediate global adoption. Importantly, the proposed motor-cognitive dual-task framework is inherently adaptable: by replacing Mahjong with culturally familiar alternatives, such as Poker, future iterations can increase cognitive variability, increase replayability, and extend applicability across diverse cultural contexts.

4 CONCLUSION

This study contributes design and empirical insights into the use of culturally contextualized AR for motor-cognitive dual-task rehabilitation with older adults. The findings indicate that the use of familiar game themes can effectively reduce perceived system complexity and support engagement in AR-based training. Meanwhile, the observed challenges related to cognitive load and hardware ergonomics highlight the importance of adaptive task design and inclusive interaction techniques when deploying AR games for aging populations.

Beyond the specific use of Mahjong, the proposed motor-cognitive dual-task framework is inherently flexible and can be adapted to alternative culturally familiar game structures, allowing broader applicability in diverse contexts. Future work will focus on refining cognitive load balance, exploring alternative interaction modalities, and conducting controlled evaluations to better understand the role of cultural grounding in long-term adoption and rehabilitation outcomes.

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